

AI Investment Update: H1 2025 Review And Strategic Outlook

Since publishing *"AI In 2025: A Roadmap for Investors"* at the onset of the year¹, developments have accelerated in many key areas. From a high level, two key takeaways have emerged YTD: (1) AI is rapidly evolving from a specialized tool into a fully democratized general-purpose technology, and (2) it's not just enhancing innovation – it's accelerating the entire innovation cycle. The scale and speed of these changes are difficult to overstate. As Neil Jacobstein, Chair of AI and Robotics at Singularity University, succinctly puts it: *"AI is moving so fast it is now a blur."*

The first half of 2025 has brought greater clarity to clients' two central questions: Where is the ROI? And where is the productivity? While hyperscaler CAPEX and monetization strategies remain nuanced, companies actively integrating AI across their operations are seeing tangible results. These gains are increasingly measurable, driven by improvements in speed, scalability, and cost efficiency – particularly in areas like customer service, research, and compliance. On the productivity front, our view remains that traditional macroeconomic indicators still struggle to capture AI's full impact. Impacts are most apparent in knowledge-intensive sectors such as law, software, healthcare, and finance, where AI enhances expert output, reduces decision latency, and enables higher-order problem-solving. Meanwhile, AI's effect on more industrial sectors (where it has a larger impact on macro data) remains

¹ Alpine Macro *Innovation Themes & Strategy* "AI In 2025: A Roadmap For Investors" (January 15, 2025).

In This Report

The General-Purpose Evolution	2
Iterative AI Improving "Reasoning"	2
The Rise Of Neoclouds	4
Long-Term Memory: Unlocking Context And Personalization.....	5
Next-Gen Silicon Chips.....	6
Key Takeaways	7

lagging in metrics. Overall, AI's integration into workflows remains uneven, with its most promising future applications still in their infancy and heavily dependent on progress in related technologies such as robotics and automation-enabling technologies.

This report provides a strategic update to our AI investment thesis, with a focus on the foundational technologies emerging as critical enablers in 2025 and their implications on the AI tech stack. While nascent, their implications span both hardware and software and are poised to drive the next wave of AI optimization. While the core themes from our initial report remain highly relevant, this update broadens our scope to include neocloud architectures, iterative AI and reasoning engines, the emergence of long-term memory, and the advancement of custom silicon. Most importantly, the confluence of these technologies is pushing AI beyond statistical pattern recognition and toward

more deliberate, task-oriented cognition – enabling systems that can reason, adapt, and learn over time. The pages ahead explore the platforms, tools, and infrastructure we believe are defining the next phase of AI’s technological maturation.

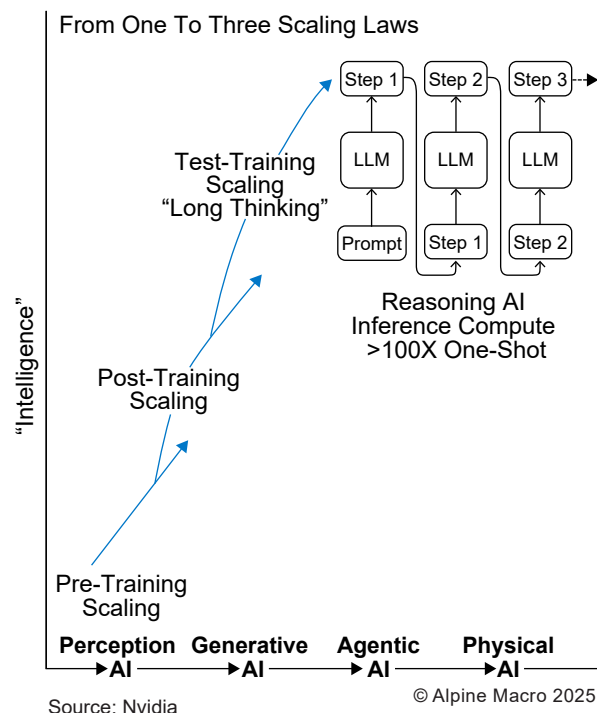
The General-Purpose Evolution

The rise of AI as a general-purpose technology (GPT) represents a foundational shift in how innovation, productivity, and economic value are created. Rooted in open-source democratization, model optimization, and developing efficiency gains, AI exhibits the three defining traits of past GPTs: rapid improvement, pervasiveness, and the ability to drive complementary innovations.

However, what sets AI apart is its unprecedented scope and self-improving nature. Unlike previous GPTs like electricity or the internet – which took decades to reshape industries through physical infrastructure – AI diffuses at digital speed which accelerates democratization. As Andrew McAfee, principal research scientist at MIT Sloan notes, generative AI is primed to spur systemic change faster than any prior GPT due to its scalability and accessibility. This drives a recursive feedback loop of accelerating capability – a phenomenon increasingly described as an “intelligence explosion”. This year alone, models with deep reasoning, multimodal inputs, and agentic roles like “Researcher” and “Analyst” have come to market alongside powerful biotech enablers – highlighting how the frontier of AI accelerates every quarter.

For investors, the implications are far-reaching as AI is not a static platform but a recursive engine

Chart 1 AI Scaling Laws Drive Exponential Demand For Compute

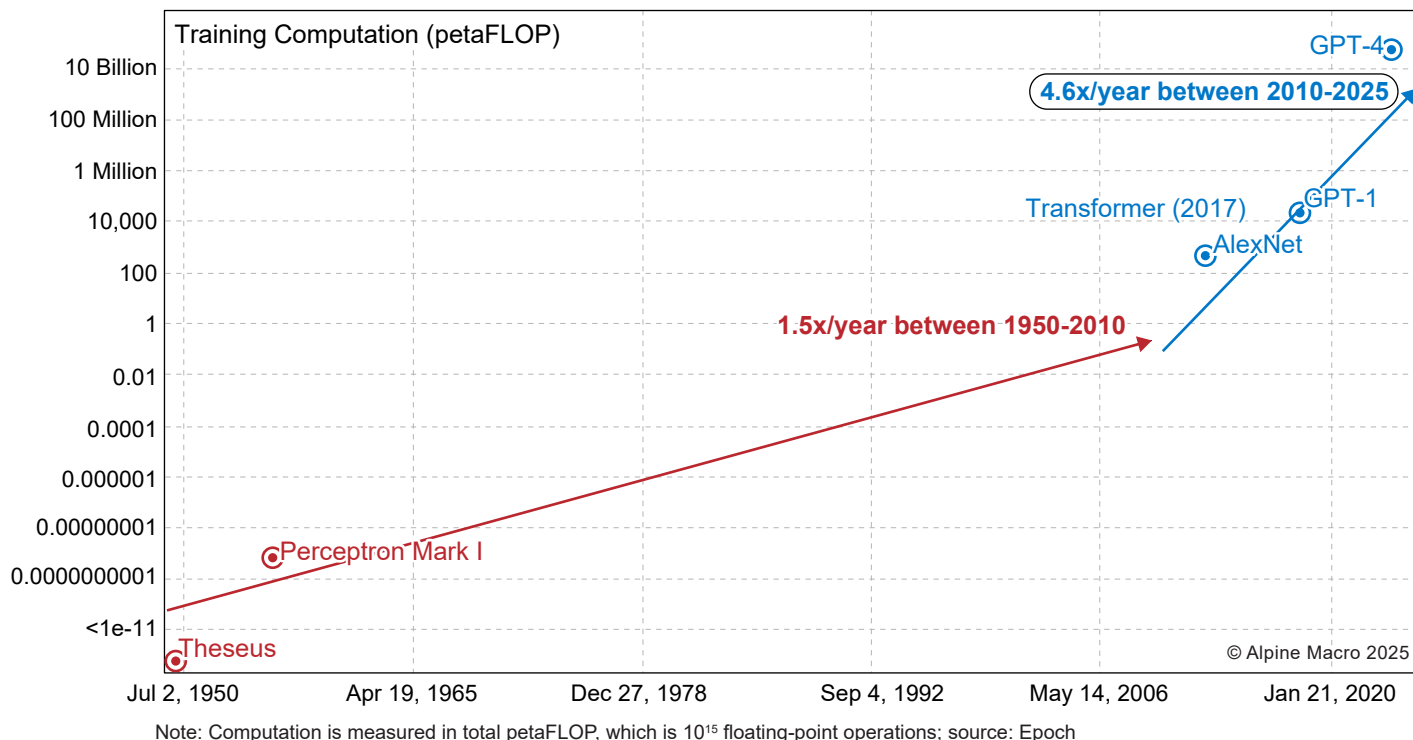


of innovation that breaks down traditional silos. Capabilities are easily diffused, allowing advances in one domain (such as drug discovery) to be rapidly applied to others (like materials science or logistics) with minimal friction. In this context, intelligence itself is becoming a commodity layer, embedded across workflows, products, and decisions.

Iterative AI Improving “Reasoning”

The proliferation of iterative AI marks a critical inflection point, advancing the field beyond statistical mimicry toward genuine machine cognition. Iterative AI are systems designed for repeated cycles of interaction, feedback, and refinement – closely mirroring how humans approach complex problems. Unlike traditional “one shot” outputs from legacy large language



Chart 2 Rapid Scaling Of Training Compute In Advanced AI Systems

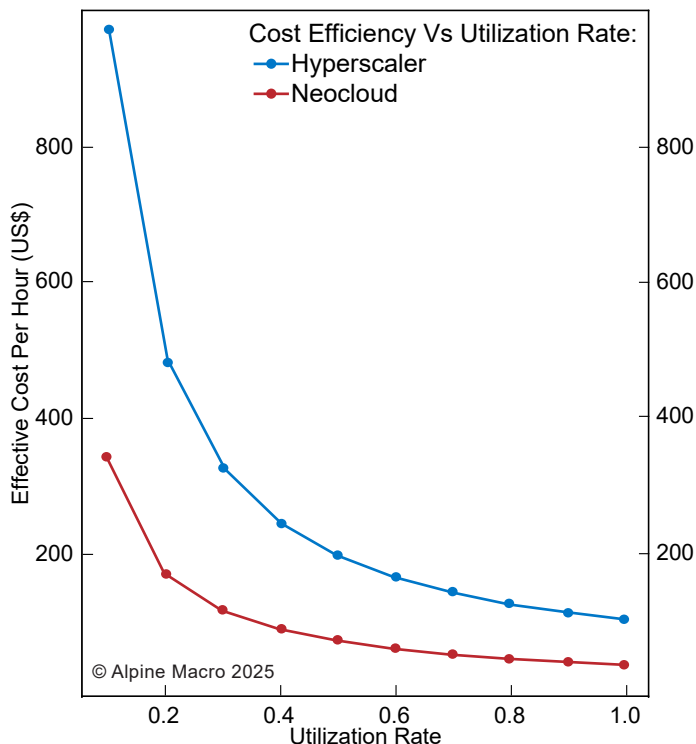
models (LLMs), iterative AI engages in cycles of challenge, refinement, and expansion through multi-step workflows that evolve dynamically. Importantly, this architecture enables multiple models to run in concert, iterating over tasks in parallel or sequence to achieve outcomes superior to what any single model could accomplish.

This shift elevates AI from a passive tool to an active thinking partner, capable of inference, context-aware reasoning, and even autonomous decision-making. While current agent-based platforms including Microsoft Copilot already demonstrate iterative functionality in tasks like multi-turn conversations, code debugging, and structured problem solving, they still require human oversight or domain-specific constraints. However, the field is advancing rapidly toward

greater autonomy, laying the groundwork for fully self-directed iterative AI agents that could independently drive projects from inception to completion.

The impact of this evolution on the AI tech stack is profound. Iterative AI's reliance on repeated cycles of computation imposes significantly higher initial demands on energy, bandwidth, and cooling compared to traditional systems due to higher compute ([Charts 1 & 2](#)). Yet, these upfront resource costs are likely to be offset by long-term efficiency gains, as unlike legacy LLMs, which typically require full retraining to incorporate new knowledge, iterative AI systems continuously learn and adapt through incremental updates, eliminating the need to retrain the entire model from scratch.

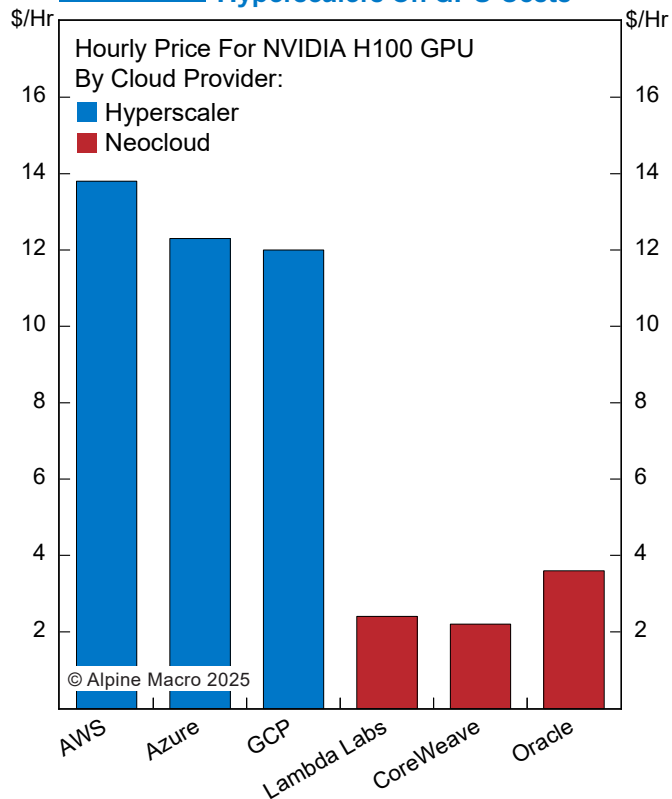


Chart 3 Strategic And Economic Drivers Of Neocloud Adoption

Source: Joseph Vazam Thomas

As AI use cases increasingly favor iteration and reasoning, demand for custom silicon (more on this below) is set to accelerate. Major cloud providers such as AWS, Google, and Microsoft are already rolling out their own specialized chips, to meet these needs. From an infrastructure perspective, iterative AI is generally more reliant on the cloud than traditional AI, given its appetite for scalable compute and storage. However, the future will likely be hybrid: cloud infrastructure will underpin the most demanding workloads, while edge computing will support latency-sensitive, privacy-critical, or real-time applications in sectors like autonomous vehicles, industrial automation, and healthcare.

2 Alpine Macro *Innovation Themes & Strategy* "Liquid Cooling: The Solution To AI's Heat And Water Problem" (October 23, 2024).

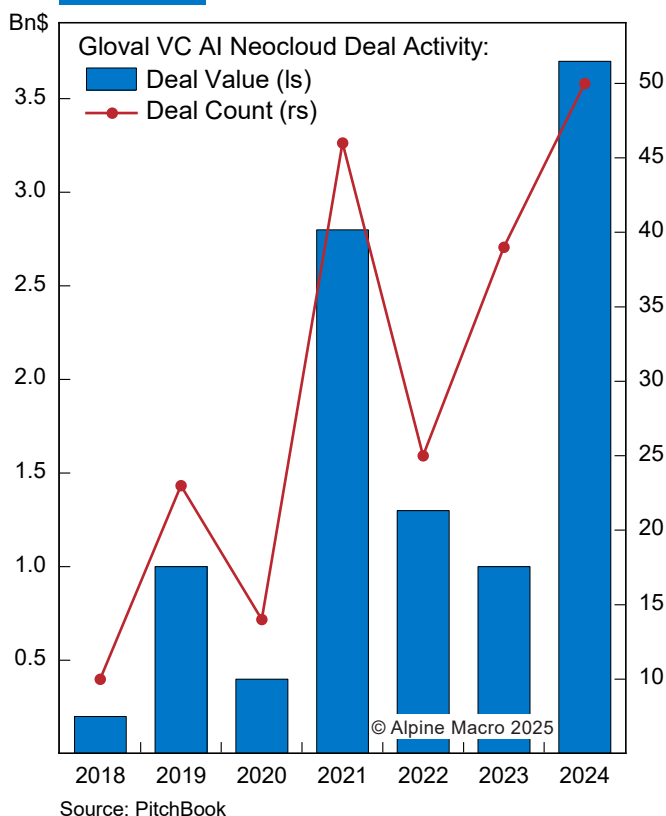
Chart 4 Neoclouds Undercut Hyperscalers On GPU Costs

Source: Joseph Vazam Thomas

The Rise Of Neoclouds

Neoclouds, a new class of AI infrastructure providers, have emerged as a critical pillar of the AI ecosystem. What sets neoclouds apart is their data center architecture, which is almost entirely built around high-density GPU clusters while intentionally minimizing the use of traditional CPUs to deliver the massive parallel processing power required for advanced AI workloads ([Charts 3 & 4](#)).

To support this, neocloud facilities leverage advanced cooling technologies – including liquid immersion and direct-to-chip cooling² – to efficiently dissipate the intense heat generated by thousands of GPUs running simultaneously. Many strategically locate their operations near regions with abundant or

Chart 5 Surge In Neocloud VC Funding

low-cost electricity. Their infrastructure also features ultra-fast, high-bandwidth networking, enabling rapid data transfer and seamless scaling for distributed AI models.

The significance of this architectural shift is highlighted through a surge in VC activity ([Chart 5](#)) and when CoreWeave went public in the largest venture-backed tech IPO since 2021, signaling both market confidence and the strategic importance of this new layer in the AI stack. Neoclouds are redefining what it means to be “AI-native”. Unlike legacy cloud or traditional machine learning workflows, neocloud platforms facilitate a more dynamic AI trajectory. This allows for rapid updates, seamless integration with other application layers, and real-time experimentation. Features such as

hot-reloading³ empower teams to iterate at the function level, validate improvements instantly, and maintain multiple model variants for different use cases. The result is a unified, evolving system where logic, memory, and tools can all adapt in response to real-world outcomes.

As demand for AI-native infrastructure accelerates, neoclouds are capturing a growing share of the market, often in partnership with leading chipmakers, including Nvidia. Notably, Nvidia has invested directly in several neocloud providers – including CoreWeave and Nebius – and has made its latest Blackwell platform first available through these partners.

Long-Term Memory: Unlocking Context And Personalization

Long-term memory is emerging as a core differentiator of generative AI systems – enabling models not just to reason and iterate, but to remember. Unlike earlier systems that operated with short-lived session context, memory-enabled AI can now retain preferences, facts, and behavior across interactions – mirroring how humans build understanding through accumulated experience. In just the first half of 2025, nearly every major AI provider has advanced in this area: Microsoft introduced persistent memory in Copilot, allowing it to recall user style and tasks; OpenAI expanded memory for ChatGPT subscribers, with user controls to review, edit, or disable memory; and competitors like xAI’s Grok, Anthropic’s Claude 4, and Google’s Gemini Advanced have followed with their own rollouts.

³ A feature that allows developers to see changes in their code reflected in a running application instantly without needing to restart or rebuild.



Technically, this capability is powered by external memory systems such as vector databases which store semantic embeddings – compressed representations of prior interactions – that are retrieved in real time to ground new responses in accumulated knowledge. On the hardware side, inference chips are being optimized to support fast memory retrieval, embedding fusion⁴, and retrieval-augmented generation (RAG)⁵ at scale. While some memory functions can be deployed at the edge, especially for privacy-critical or latency-sensitive applications, most long-term memory use cases still rely heavily on centralized cloud resources, boosting demand for memory-intensive data center capacity.

Persistent memory unlocks more personalized, efficient, and anticipatory AI – transforming customer support, research, creative workflows, and software agents. Over time, this gives rise to niche models with a form of digital identity – systems that not only “know” facts but also “know” you: how you think, write, prioritize, and decide. In this context, memory becomes a strategic asset, and context and continuity represent the next key technological “moats”.

Next-Gen Silicon Chips

Custom silicon is enabling purpose-built chips that deliver greater efficiency, lower power consumption, and tailored performance for specific AI workloads. Unlike general-purpose GPUs such as Nvidia's Blackwell – which are designed for broad applicability and excel at high raw performance – custom silicon is engineered for specialized tasks, allowing

companies to optimize for massive parallelism, low latency, and high memory bandwidth. Examples like AWS's Trainium3, Google's Trillium, and Microsoft's Maia 100 illustrate how cloud providers are building proprietary chips to support advanced AI workflows, including agentic reasoning, continuous learning, and persistent memory.

The key distinction between custom silicon and Nvidia's Blackwell lies in specialization and control. Custom chips are optimized for particular workloads and can integrate features like enhanced memory management and security, making them ideal for both cloud and edge deployments where efficiency and context matter most. We note this specialization could open the door for hyperscalers to reduce reliance on Nvidia for niche applications, fine-tune their infrastructure, and better align hardware with evolving AI models and business needs. However, Blackwell remains the gold standard for general-purpose compute, offering unmatched flexibility and performance for the largest, most demanding AI models.

Ultimately, custom silicon enables the next phase of AI by making persistent, context-aware, and domain-specific models more technically and economically viable. This marks a strategic shift away from one-size-fits-all GPUs toward a more diversified, specialized, and strategically controlled AI hardware ecosystem – empowering more personalized AI experiences at scale and at the edge.

4 The process of combining embeddings from different sources or modalities (like text, images, or sensor data).

5 RAG allows LLMs to access and incorporate information from external data sources (like documents, databases, or the internet).

Key Takeaways

The first half of 2025 has confirmed that AI is a fully democratized, general-purpose technology. As intelligence becomes a ubiquitous layer embedded in workflows and products, strategic advantage is shifting toward those who can best integrate, personalize, and scale AI-driven solutions. This represents a diversification opportunity for companies like Nvidia to provide an “all-encompassing” tech stack across hardware and software.

Value creation is shifting away from generic models to domain-specific systems fine-tuned for real-world constraints. This transition toward specialization tailoring AI to specific industries, functions, or workflows marks a critical inflection for stakeholders. Rather than chasing model size, albeit remaining a key area of focus in the long-term for AGI, strategic focus is evolving into delivering lean, fine-tuned systems that outperform through specialization and contextual intelligence.

As a result, at the technology stack level, foundational shifts are underway that will define the next wave of true AI value. The convergence of neocloud architectures, iterative AI, long-term memory, and custom silicon is pushing AI beyond pattern recognition toward true reasoning and adaptive cognition. They are creating a new layer of demand for infrastructure, hardware, and orchestration tools built for this more complex, performance-sensitive era. In this cycle, the leading investment opportunities lie at the intersection of enabling technologies and applied AI, as the market shifts from experimentation to measurable ROI and broad-based

productivity gains. The winners will be those who emerge as hardware and software leaders, offering complementary solutions designed to work in tandem and unlock tangible value across sectors.

Noah Ramos

Global Strategist

EDITORIAL BOARD

Noah Ramos

Global Strategist

Ria Nagpal

Research Analyst

Chen Zhao

Chief Global Strategist

David Abramson

Chief U.S. Strategist &
Director of Research



Disclaimer and copyright restrictions © Alpine Macro 2025. All rights reserved.

The information, recommendations, analysis and research materials presented in this document are provided for information purposes only and should not be considered or used as an offer or solicitation to sell or buy financial securities or other financial instruments or products, nor to constitute any advice or recommendation with respect to such securities, financial instruments or products. This document is produced for subscribers only, represents the general views of Alpine Macro, and does not constitute recommendations or advice for any specific person or entity receiving it. The text, images and other materials contained or displayed on any Alpine Macro products, services, reports, emails or website (including this report and its contents) are copyrighted materials proprietary to Alpine Macro and may not be circulated without the expressed authorization of Alpine Macro. If you would like to use any graphs, text, quotes, or other material, you must first contact Alpine Macro and obtain our written authorization. Alpine Macro relies on a variety of data providers for economic and financial market information. The data used in this publication may have been obtained from a variety of sources including Bloomberg Finance L.P., Macrobond, CEIC, Choice, MSCI, BofA Merrill Lynch and JP Morgan. The data used, or referred to, in this report are judged to be reliable, but Alpine Macro cannot be held responsible for the accuracy of data used herein.